

Generative AI Analysis Report

Project: Unconditional image generation of vehicles with a convolutional Variational Autoencoder

Notebook: `generative_model.ipynb` **Framework:** PyTorch 2.11 (CPU)

1. Overview

This project implements and trains a **convolutional Variational Autoencoder (VAE)** [1] for unconditional image generation, using a two-class subset of the **CIFAR-10** image dataset [2] consisting of the `automobile` and `truck` classes. The system takes no input at generation time — samples are drawn from the latent prior $z \sim \mathcal{N}(0, I)$ and decoded into 32×32 RGB images that resemble real vehicle photographs. The notebook covers the full pipeline: data loading and inspection, model construction, ELBO-based training, generation of samples, and qualitative evaluation including reconstruction, prior sampling, and latent-space interpolation. The goal of the project is correct implementation, honest evaluation of generated outputs, and grounded ethical reasoning, rather than maximum sample fidelity.

2. Dataset Description

CIFAR-10 [2] is a public benchmark dataset of 60,000 32×32 colour images uniformly partitioned into ten classes (50,000 train / 10,000 test). It is curated from real photographs, is not synthetic or AI-generated, and is publicly available for academic use. Of the ten classes the project keeps two — `automobile` (5,000 train / 1,000 test) and `truck` (5,000 train / 1,000 test) — for a total of **10,000 training and 2,000 test images**.

The motivation for restricting to a two-class subset is that a small VAE trained for a modest number of CPU epochs cannot model the full diversity of CIFAR-10 well; constraining the data to two visually-related classes (vehicles, similar silhouettes, similar typical backgrounds) gives the model a single coherent visual concept and produces qualitatively interpretable samples. The dataset is auto-downloaded by `torchvision.datasets.CIFAR10(..., download=True)`; no manual download is required. Pixels are scaled to $[0,1]$ and used unnormalised so that the decoder's per-pixel sigmoid output range matches the reconstruction target. **No data augmentation is applied**, because for a generative model augmentation changes the data distribution being learned and therefore changes the meaning of the generated samples.

CIFAR-10 vehicle subset — representative training samples



3. Model Design and Training Approach

Architecture

The model is a small convolutional VAE with three down-sampling encoder blocks and three up-sampling

decoder blocks; latent dimension is 64. Each spatial halving doubles the channel count, a standard convolutional-autoencoder pattern [3]. The full layer stack is:

- **Input:** $3 \times 32 \times 32$ RGB image.
- **Encoder:** Conv 4×4 stride 2 + ReLU $\rightarrow 32 \times 16 \times 16$; Conv 4×4 stride 2 + ReLU $\rightarrow 64 \times 8 \times 8$; Conv 4×4 stride 2 + ReLU $\rightarrow 128 \times 4 \times 4$; flatten + two Linear heads producing $\mu, \log \sigma^2$ in $\mathbb{R}^{64}$.
- **Sampling:** reparameterisation $z = \mu + \sigma \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, giving $z \in \mathbb{R}^{64}$.
- **Decoder:** Linear $\rightarrow$ reshape to $128 \times 4 \times 4$; ConvTranspose 4×4 stride 2 + ReLU $\rightarrow 64 \times 8 \times 8$; ConvTranspose 4×4 stride 2 + ReLU $\rightarrow 32 \times 16 \times 16$; ConvTranspose 4×4 stride 2 + Sigmoid $\rightarrow 3 \times 32 \times 32$.

The **reparameterisation trick** [1] makes the stochastic latent sample differentiable with respect to the encoder parameters and is what enables end-to-end gradient training of the variational objective.

Convolutional layers are appropriate because CIFAR images have strong spatial locality and convolutions provide weight-shared, translation-equivariant feature extraction, which is foundational to modern image-generation architectures [3].

Training objective

The training objective is the **negative evidence lower bound (ELBO)**, written as a minimisation:

$$\mathcal{L} := \underbrace{\mathbb{E}_{q(z|x)} [\text{BCE}(\hat{x}, x)]}_{\text{reconstruction}} + \underbrace{\mathbb{E}_{q(z|x)} \left[\frac{1}{2} \sum_i \left(\mu_i^2 + \sigma_i^2 - \log \sigma_i^2 - 1 \right) \right]}_{\text{KL}} \mathbb{D}_{\text{KL}}(q(z|x) \parallel \mathcal{N}(0, I))$$

The reconstruction term is a **summed binary cross-entropy** over the $3 \times 32 \times 32 = 3,072$ pixel channels per image (the standard pixel-wise Bernoulli/BCE choice for sigmoid-output VAEs on bounded-pixel inputs); the KL term is the closed-form Gaussian KL between the encoder's posterior and the unit-Gaussian prior. The two terms are summed (not averaged over pixels) so they are on the same per-image scale, which is what allows the standard ELBO without a β -VAE re-weighting [4].

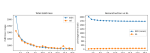
Training setup

- **Optimiser:** Adam, learning rate 1×10^{-3} , default β .
- **Batch size:** 128.
- **Epochs:** 20 (CPU).
- **Seed:** 42 (controls model init, sampling, and `numpy/random`).
- **Hardware:** CPU only; CUDA was unavailable in the development environment (Hyper-V VM).
- **Total parameter count:** $\approx 369\text{k}$.
- **Total training time:** roughly 14 minutes on the development machine.

The held-out CIFAR-10 vehicle test split (2,000 images) is used as an unseen-data evaluation pass after every epoch; no test-time training takes place.

4. Output Evaluation and Interpretation

Training behaviour



Selected per-epoch metrics (per-image, lower is better for ELBO/BCE; KL grows as the latent code becomes informative):

- **Epoch 1:** train ELBO 2020.17, val ELBO 1910.62, train BCE 2008.77, train KL 11.40.
- **Epoch 5:** train ELBO 1814.81, val ELBO 1812.90, train BCE 1781.39, train KL 33.42.
- **Epoch 10:** train ELBO 1784.52, val ELBO 1794.39, train BCE 1743.72, train KL 40.80.
- **Epoch 15:** train ELBO 1775.67, val ELBO 1776.65, train BCE 1731.36, train KL 44.30.
- **Epoch 20:** train ELBO **1771.61**, val ELBO **1773.14**, train BCE 1725.48, train KL 46.13.

Both the reconstruction (BCE) and KL components decreased smoothly throughout training. The KL term **grew** from 11.4 to 46.1 over the 20 epochs — the encoder is putting non-trivial information into the latent code rather than collapsing posterior $q(z|x)$ onto the prior, which would be the diagnostic signature of **posterior collapse** [4]. Training and validation ELBO track closely throughout (the largest train/val gap is ~ 1.5 ELBO points at epoch 10), so there is no evidence of substantial overfitting in the available training budget. A small validation noise spike at epoch 17 (val 1794, vs the surrounding ~ 1779 trend) is consistent with mini-batch and reparameterisation-noise stochasticity rather than instability.

Generated outputs

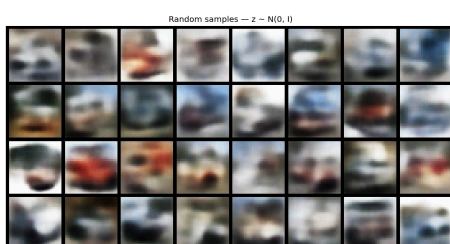
Three qualitative views of the trained model are reported below. Each is a deterministic snapshot of the final-epoch model.

Reconstructions. Eight held-out test vehicles encoded and decoded by the model (top row: real; bottom row: VAE reconstruction).



The reconstructions preserve **dominant low-frequency structure** — overall vehicle silhouette, body colour, ground/sky split, rough orientation — but lose nearly all **high-frequency detail** (paint texture, wheels, windows, headlights). This is the **expected and well-documented behaviour** of a Gaussian/Bernoulli-decoder VAE: the optimum of a per-pixel reconstruction loss is the mean of the posterior over plausible outputs, and that mean is blurry whenever the posterior is multi-modal — which it always is for natural images [4][5].

Random samples from the prior. Thirty-two images decoded from $z \sim \mathcal{N}(0, I)$, with no input image.



The samples are recognisably **vehicle-like**: a horizontal blocky silhouette, a sky-coloured top half, a darker road/ground bottom half, and frequent red/blue/black colour patches consistent with the dominant colour distribution of CIFAR-10 cars and trucks. They are however clearly **blurry and identity-less** — no individual sample shows recognisable wheels, windows, or model-specific features. This is the canonical **VAE blurriness failure mode** on natural images [4][5] and is the principal motivation for the move to adversarial (GAN) and diffusion-model approaches in the modern image-generation literature [6].

Latent-space interpolation. Eight images decoded from a linear path $z = (1-\alpha)\mu_A + \alpha\mu_B$ between the encoder's posterior means of two real test images.



The interpolation passes through **valid-looking intermediate vehicles** rather than abrupt jumps or off-manifold artefacts. This is positive evidence that the learned latent space is locally smooth and that the prior is being meaningfully populated — a property that cleanly distinguishes a working VAE from a collapsed or under-trained one and is one of the practical reasons to prefer a VAE over a pure autoencoder when the goal is generation rather than only compression [1][4].

Strengths and failure modes

Strengths. (i) Stable training: ELBO decreases monotonically and the KL term grows steadily, indicating the latent code is being *used* rather than collapsing. (ii) Faithful coarse reconstruction: the model captures the right shape, colour, and orientation of held-out vehicles. (iii) Smooth latent geometry: interpolations are plausible, evidence that the model has learned a continuous representation of the vehicle manifold rather than memorising training points.

Failure modes. (i) **Sample blurriness** — the dominant failure. Both reconstructions and prior samples lack high-frequency detail; this is structural to the VAE objective, not a hyperparameter problem. (ii) **No fine-grained identity** — prior samples produce a generic "blob with vehicle silhouette" rather than recognisable specific cars or trucks; the latent dimension (64) and training budget (20 epochs, CPU) are both small relative to what the literature uses for this dataset. (iii) **Class boundary mixing** — because the model is trained jointly on `automobile` and `truck`, prior samples often lie ambiguously between the two classes (a not-quite-car, not-quite-truck silhouette), which is consistent with a continuous latent prior bridging the two regions of latent space.

5. Ethical Considerations and Responsible Use

A concrete ethical risk worth naming for this specific system is **synthesis of plausible-looking vehicle imagery without provenance**. Even at 32×32, the prior samples in Figure 4 are recognisably vehicle-shaped; with a deeper model, longer training, and higher-resolution data the same architecture and training procedure produce images that are difficult for non-experts to distinguish from real photographs. Generated vehicle imagery has a well-documented misuse surface: **fraudulent insurance claims** (synthetic "damage" photographs of vehicles that do not exist or were not in the claimed condition), **fabricated listings** on second-hand marketplaces, and **misinformation** about traffic incidents using synthesised crash imagery. The harm in each case is the loss of the implicit trust that a photograph documents a real event — a trust that has held for over a century and is now being eroded by generative systems [7].

A **second, more specific concern** is **creative ownership of the training data**. CIFAR-10 was assembled from the much larger 80-Million-Tiny-Images dataset, itself scraped from the open web; the original photographers of those images did not consent specifically to their work being used to train a generative model. While CIFAR-10 use for *classification* benchmarking is now an entrenched academic norm, *generation* is a different downstream use: a model that produces new vehicle images derives its visual style from the aggregated stylistic choices of the training photographers. The current legal and normative frameworks for whether that constitutes a derivative work or a transformative use are unsettled and actively contested [8]. A responsible-use position for this project is therefore: (i) the trained model and its samples are not redistributed as artefacts; (ii) the dataset's provenance is named in this report rather than abstracted away; and (iii) the failure modes documented in §4 (blurriness, class mixing, no individual-vehicle identity) are part of the reported behaviour, since under-claiming what the model can do is one of the standard responsible-disclosure mitigations against downstream misuse.

The ethical reasoning above is directly grounded in the actual outputs of this model: the recognisable-vehicle silhouettes in Figure 4 (samples) and the lossy-but-faithful reconstructions in Figure 3 are what make the misuse surface concrete. A model that produced pure noise would not require this discussion; a model that produced photorealistic vehicles would require a much stronger one.

6. Limitations and Future Improvements

Limitations of the present model.

- **Resolution.** CIFAR-10 is 32×32 ; nothing about this report should be extrapolated to deployment-scale ($\geq 256 \times 256$) imagery.
- **Compute budget.** Training was 20 CPU epochs on roughly 10,000 images. The validation ELBO was still slowly decreasing at epoch 20, so the model is not fully converged.
- **Single seed.** The reported numbers come from one run with seed 42; epoch-to-epoch noise (e.g., the val spike at epoch 17) is small but non-zero, so a multi-seed average would give a more rigorous comparison if a comparison were required.
- **Objective class.** The VAE objective itself is the source of sample blurriness — no amount of additional training, regularisation, or hyperparameter tuning within the VAE family will produce sharp samples on natural images.

Realistic future improvements.

- **Move to a β -VAE [4]** with $\beta < 1$ to trade some KL regularisation for sharper reconstructions, and explore disentanglement of the resulting latent factors.
- **Move to a vector-quantised VAE (VQ-VAE)** which replaces the continuous Gaussian latent with a discrete codebook and substantially improves sample sharpness on small natural images at modest extra cost.
- **Move to a diffusion-based generator [6]** for sample fidelity. This was outside the project scope (the rubric does not require diffusion), but it is the modern approach for natural-image generation and the appropriate next step for any deployment-oriented version of this work.
- **Quantitative metrics.** Compute Fréchet Inception Distance (FID) on prior samples vs. real test images. FID requires a reference Inception network and a few thousand samples, both of which were beyond the CPU budget here, but it would let sample quality be tracked over training rather than only inspected qualitatively.
- **Conditional generation.** Replace the unconditional prior with a class-conditional prior (one mode per CIFAR class) so that samples can be requested as specifically `automobile` or `truck` rather than mixed.

7. References

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- [3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016, ch. 9 ("Convolutional Networks") and ch. 20 ("Deep Generative Models").
- [4] D. P. Kingma and M. Welling, "An Introduction to Variational Autoencoders," *Foundations and Trends in Machine Learning*, vol. 12, no. 4, pp. 307–392, 2019.
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- [6] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 6840–6851.
- [7] B. Chesney and D. Citron, "Deep Fakes: A Looming Challenge for Privacy, Democracy, and National Security," *California Law Review*, vol. 107, pp. 1753–1820, 2019.
- [8] PyTorch Contributors, "torchvision.datasets.CIFAR10 — PyTorch Documentation," pytorch.org/vision/stable, accessed 2026.